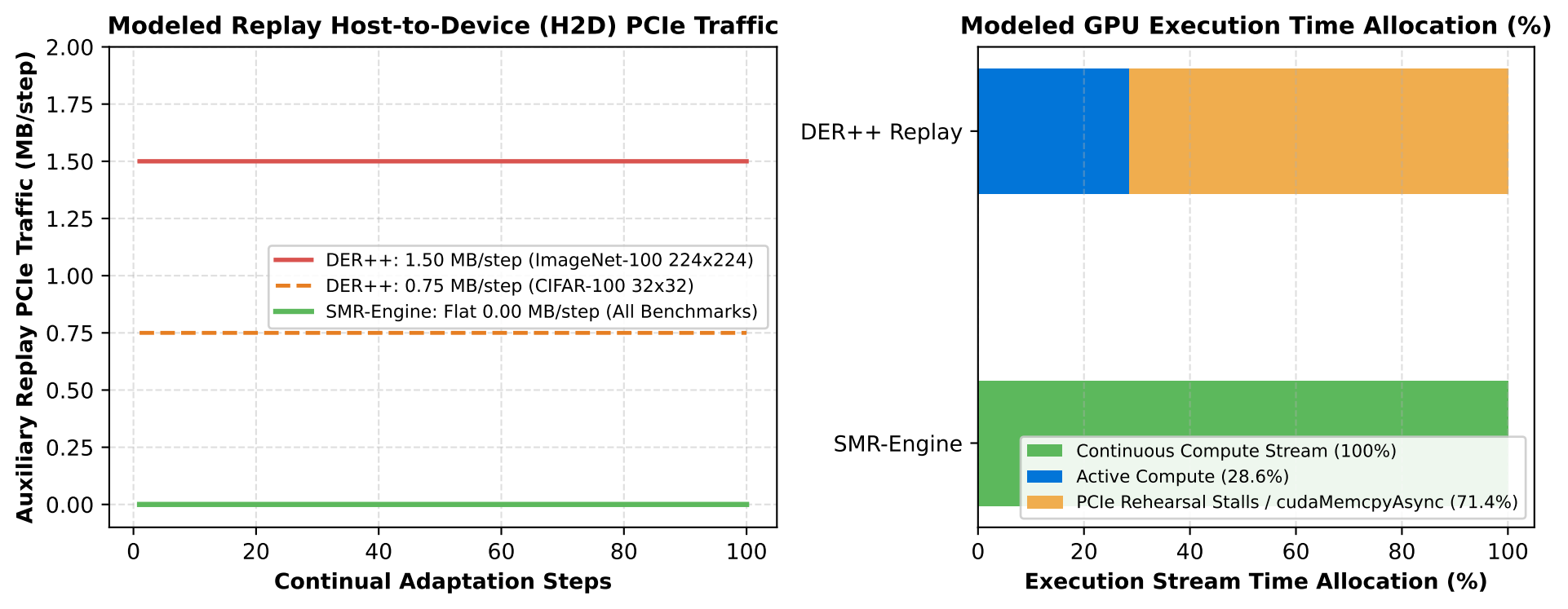
Modeled Replay Host-to-Device (H2D) PCIe Traffic
Auxiliary Replay PCIe Traffic (MB/step)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0
20
40
60
80
100
Continual Adaptation Steps
DER++: 1.50 MB/step (ImageNet-100 224x224)
DER++: 0.75 MB/step (CIFAR-100 32x32)
SMR-Engine: Flat 0.00 MB/step (All Benchmarks)
Modeled GPU Execution Time Allocation (%)
DER++ Replay
SMR-Engine
Continuous Compute Stream (100%)
Active Compute (28.6%)
PCIe Rehearsal Stalls / cudaMemcpyAsync (71.4%)
0
20
40
60
80
100
Execution Stream Time Allocation (%)